



VISION: Mission

SLSU shall provide global education that nurtures professionals who are morally grounded and responsive to community needs. The university is also committed to conduct research and extension across various fields while innovatively utilizing and regenerating its resources through sustainable environmental approaches.

MISSION:

A globally recognized university known for academic excellence, innovative and responsive research and extension services for national and global development.

CORE VALUES:

- Go** – God-loving
- S** – Service-oriented
- L** – Leadership by Example
- S** – Sustained Passion of Excellence
- U** – Undiminished Commitment to Peace and Environment Advocacy.

COLLEGE OF ENGINEERING GOALS:

1. To develop committed competitive, competent and ethical professionals in the field of Engineering.
2. To prepare students for their career as engineers by providing relevant and responsive circular program as dedicated and proficient faculty and advanced facilities.
3. To educate students good moral values, entrepreneurial capability and sense of responsibility in sustaining a healthy environment.

Program Educational Objectives		College of Engineering Goals		
		1	2	3
1	To equip graduates with technopreneurship skills with professional ethics and to inculcate the value of the continuance of professional development based on the importance of lifelong learning.	✓	✓	✓
2	To provide knowledge that are based on system analysis, system design & development and project management; and to espouse research & development that are current in content, cutting edge and state-of-the-art in delivery and	✓	✓	✓

Course Title: Embedded Systems	Date Revised: August 2025	Date Effective: August 11, 2025	Prepared by: Engr. Dean Adrian L. Nombrefia	Checked by: Engr. Julie Ann B. Susa-Gili	Approved by: Dr. Maria Corazon B. Abejo	Page 1 of 6
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	responding to the bio local & global technological challenges of the times.			
3	To inculcate to the students the importance of healthy environment for the people geared with responsibility equipped with knowledge in science & technology and environmental values.	✓	✓	✓

COURSE SYLLABUS

1. Course Code
: CPE24
2. Course Title
: Embedded Systems
3. Pre-requisite
: Microprocessors
4. Co-requisite
: Embedded Systems Laboratory
5. Credit/ Class Schedule
: 3 units Lecture (3 Hours per Week)
TTH - 9:00 – 10:30 / 10:30 – 12:00
6. Course Description
: This course provides advanced topics in embedded systems design using contemporary practice; interrupt-driven, reactive, real-time, object- oriented, and distributed client/server embedded systems.

7. Program Outcomes and Relationship to Program Educational Objectives:

Program Outcomes		Program Educational Objectives		
		1	2	3
a	Ability to apply knowledge of mathematics and science to solve engineering problems;	✓	✓	✓
b	An ability to design and conduct experiments as well as analyze and interpret data.	✓	✓	✓
c	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards;	✓	✓	✓
d	Ability to function on multidisciplinary teams;	✓	✓	✓
e	Ability to identify, formulate, and solve engineering problems;	✓	✓	✓
f	Understanding of professional and ethical responsibility;	✓	✓	✓
g	Ability to communicate effectively;	✓	✓	✓
h	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	✓	✓	✓

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i	Recognition of the need for, and an ability to engage in life-long learning	✓	✓	✓
j	Knowledge of contemporary issues.	✓	✓	✓
k	Ability to use techniques, skills, and modern engineering tools necessary for engineering practice; and	✓	✓	✓
l	Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments.	✓	✓	✓

8. Course Objectives and Relationship to Program Outcomes:

Course Objectives		Program Outcomes											
		a	b	c	d	e	f	g	h	i	j	k	l
1	Analyze embedded system concepts and application		✓	✓									
2	Distinguish which embedded system is appropriate to use in solving engineering problems.		✓	✓									
3	Evaluate different embedded systems and its components and application		✓	✓									
4	Develop an application of embedded systems		✓	✓									

9. Course Outline and Timeframe:

Week	Topics
1	Introduction to Embedded Systems & Course Preliminaries
2	History and Overview
3	Relevant Tools, Standards, and/or Engineering Constraints
4	Characteristics of Embedded Systems
5	Basic Software Techniques for Embedded Applications
6	Preliminary Examination Period
7	Parallel Input and Output
8	Asynchronous and Synchronous Serial Communication
9	Periodic Interrupts, Waveform Generation, Time Measurement
10	Data Acquisition, Control, Sensors, and Actuators part 1
11	Data Acquisition, Control, Sensors, and Actuators part 2
12	Midterm Examination Period
13	Implementation Strategies for Complex Embedded Systems
14	Techniques for Low-Power Operation
15	Mobile and Networked Embedded Systems
16	Advanced Topics on Input/Output
17	Computing Platforms for Embedded Systems
18	Final Examination Period

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10. Required Readings:

Jimenez, Manuel, Palomera, Rogelio and Couvertier, Isidro. Introduction to Embedded Systems using microcontrollers and the MSP430

11. Suggested Readings and References:

Marwedel, Peter. Embedded System Design Embedded Systems Foundations of Cyber-Physical Systems, and the Internet of Things Fourth Edition

12. Course Requirements:

- Attendance and Recitation
- Assignment and Seatwork
- Quizzes
- Examinations

13. Grading System:

In order for a student to pass the subject, he/she must:

- Attend to class regularly and on time;
- Accomplish Laboratory Exams, Term Lab Test and Laboratory Exercises;

Passing Rate: 60 %

Grading System:

- Performance Output – 40%
- Term Examinations – 30%
- Recitation – 15%
- Quizzes – 15%

Aside from academic deficiency, other ground for a failing grade is more than 20% of the total number of attendances missed.

14. Classroom Policies:

1. Every student is expected to attend classes on time.
2. Every student must wear proper uniform during classes.
3. Every student must accomplish Seatwork, Quizzes and Examinations completely and on time.
4. For Recitations, Reports and Assignments, the student must comply and perform accordingly during classes.
5. The class should abide the classroom policies given by the College.

15. Consultation Hours:

Tuesday, Thursday and Friday 3:00 - 4:30

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16. Course Outcomes and Relationship to Course Objectives:

Course Outcomes		Course Objectives			
A student completing this course should at the minimum be able to:		1	2	3	4
1	Recognize and distinguish different embedded systems and applications.	✓	✓	✓	✓
2	Analyze what is the best embedded systems is appropriate in solving engineering problems.			✓	✓
3	Create a fully functioning embedded applications.			✓	✓

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17. Course Learning Plan

Learning Outcome	Topic	Methodology	Resources	Assessment
Recognize and distinguish different embedded systems and applications.	History and Overview Relevant Tools, Standard and/or Engineering Constraints Characteristics of Embedded Systems Basic Software Techniques for Embedded Applications	Discussion of given topic	Module/ Handouts, Power Point Presentation	Assignment, Presentation, Recitation, Seatwork, Quizzes, Examination, TOS
Analyze what is the best embedded systems is appropriate in solving engineering problems.	Parallel Input and Output Asynchronous and Synchronous Serial Communication Periodic Interrupts, Waveform Generation, Time Measurement Data Acquisition, Control, Sensors, and Actuators	Discussion of given topic	Module/ Handouts, Power Point Presentation	Assignment, Presentation, Recitation, Seatwork, Quizzes, Examination, TOS
Create a fully functioning embedded applications.	Implementation Strategies for Complex Embedded Systems Techniques for Low-Power Operation Mobile and Networked Embedded Systems Advanced Topics on Input/Output Computing Platforms for Embedded Systems	Discussion of given topic	Module/ Handouts, Power Point Presentation	Assignment, Presentation, Recitation, Seatwork, Quizzes, Examination, TOS

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